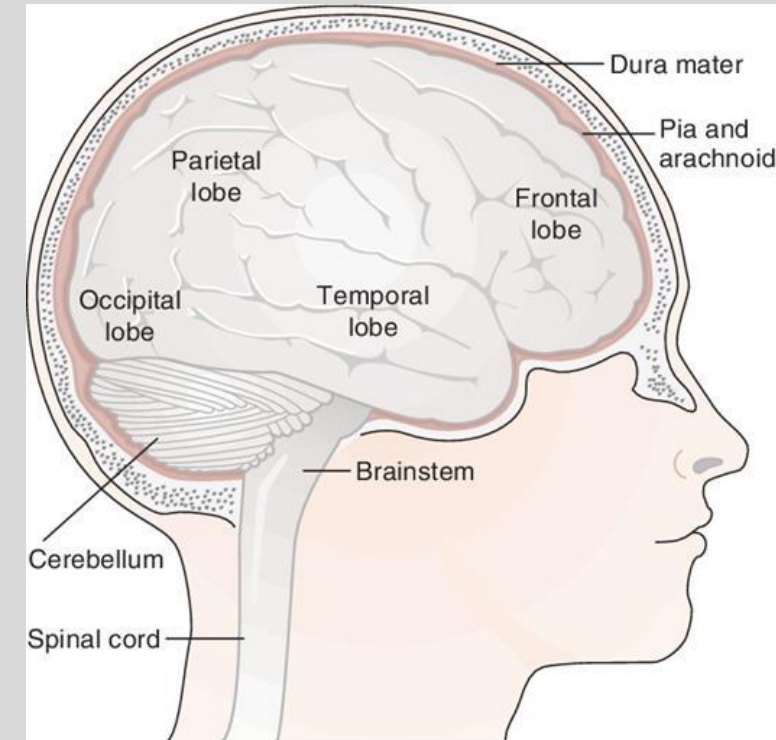


Inflammation of the CNS

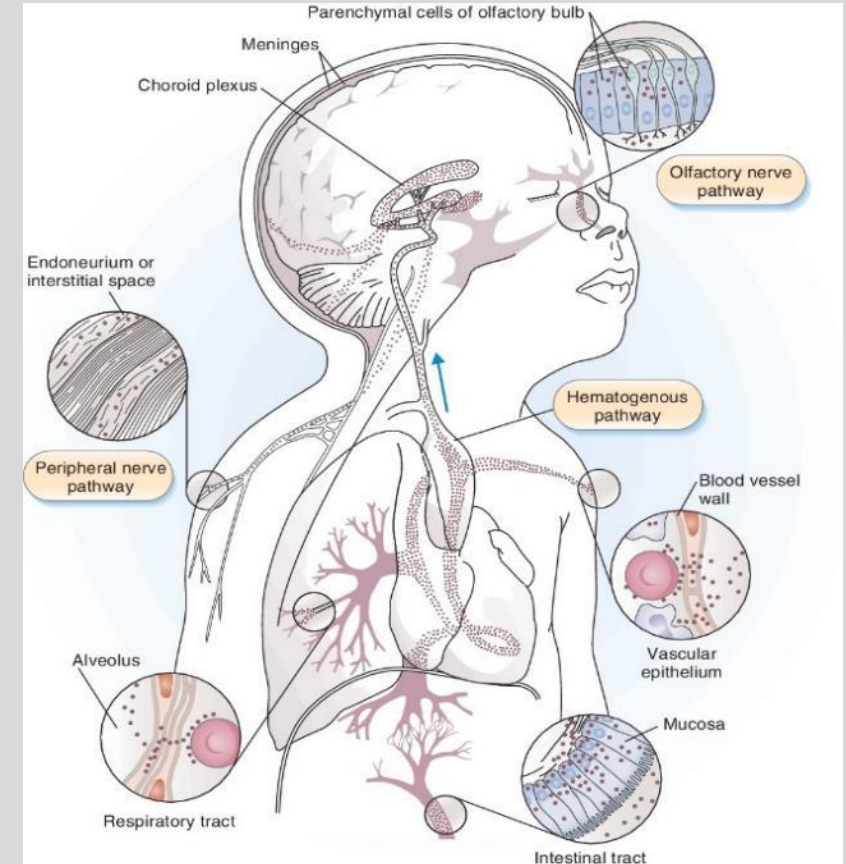
(review)

- Meningitis (meninges)
 - Acute or chronic
 - Bacterial, viral, fungal, or aseptic
 - Encephalitis (parenchyma)
 - Brain abscesses
 - Myelitis (spinal cord)
-
- *Viruses mainly cause acute meningitis & encephalitis*



How CNS Infection Occurs

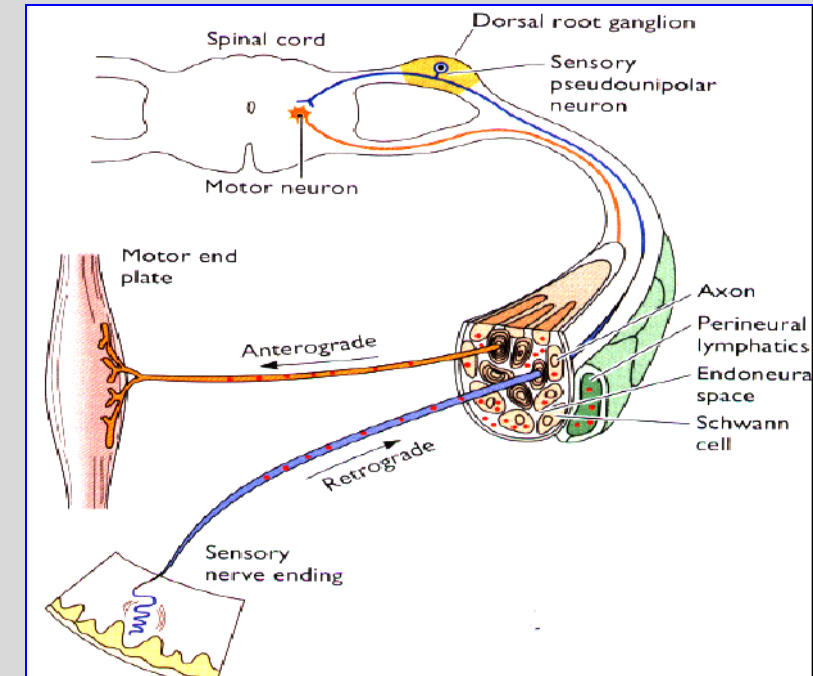
- Entry to the CNS
 - Hematogenous from inoculation at another site
 - Respiratory epithelium (aerosol, droplets; e.g. meningococci)
 - Animal bite (rabies virus), or bug bites (West Nile virus & other arboviruses)
 - Transplacentally (cytomegalovirus, rubella virus or Zika virus).
 - **from blood to CNS: choroid plexus or infection of CNS endothelial cells
- Direct inoculation
 - Trauma
 - Surgery



How Virions reach the CNS

- Neurotropism of virions

- Poliovirus for motor neurons of the spinal cord and medulla
- Mumps virus for ependymal cells lining the ventricles in the fetal brain
- Rabies virus enters the distal terminals of peripheral nerves and travels inside nerve axons via axoplasmic transport to reach nerve cell bodies in sensory ganglia and the spinal cord.
- Some viruses and other infectious agents can penetrate the body through olfactory nerve endings (direct contact with the external environment)
 - a) Nasal infections occur with rabies virus and some arboviruses under conditions with high airborne concentrations of virus, as in caves filled with infected bats.



How Virions reach the CNS

- Neurotropism of virions
 - Following chickenpox, which results from primary infection with varicella-zoster virus (VZV), virus spreads from the skin through sensory nerves to become latent in neurons in sensory ganglia. Reactivation results in spread of virus through sensory nerves to produce zoster (“shingles”)
 - HSV spreads through nerves from sites of primary infection such as oropharyngeal (primarily HSV-1) or genital mucosa (primarily HSV-2) to infect trigeminal or sacral ganglia
 - a) Virus becomes latent in sensory neurons.
 - b) Reactivation results in spread of virus along sensory nerves to mucosal surfaces, producing recurrent herpes labialis (cold sores) or herpes genitalis.

Epidemiology

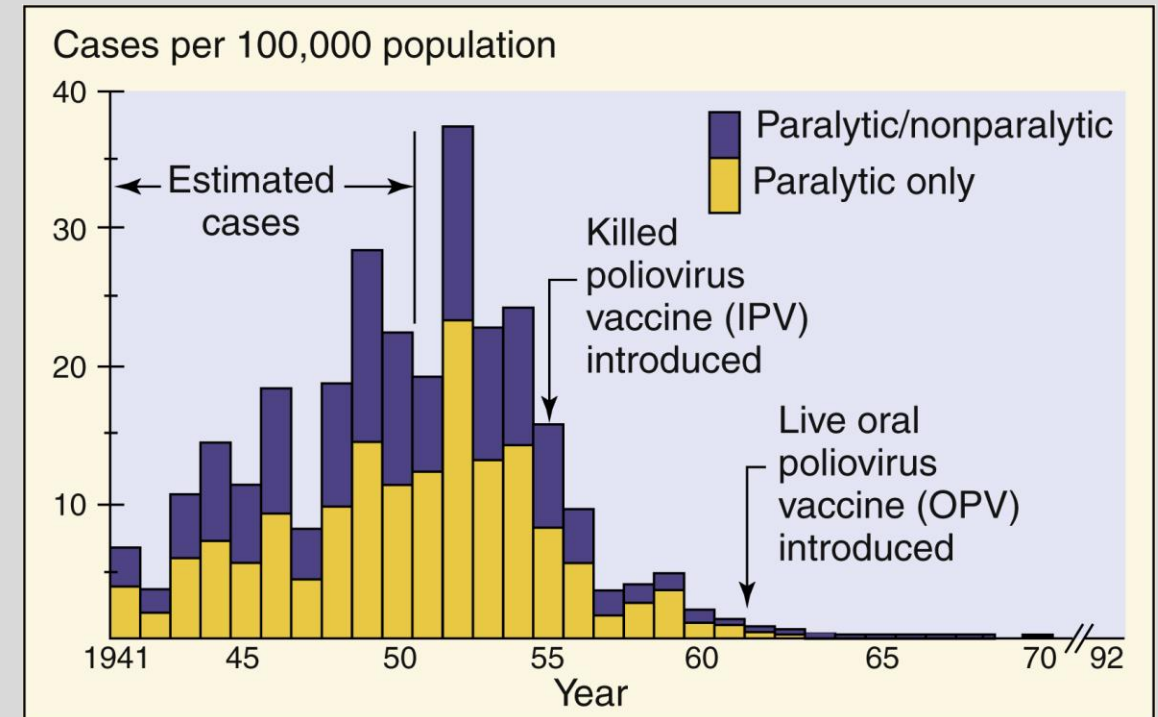
- Seasonal: e.g. arboviruses spread by vectors (mosquito and ticks) more commonly in summer
- Enterovirus infections usually peak in the late summer and fall.
- Other pathogens don't have specific seasonal variation and occur with equal frequency throughout the year; e.g. HSV (most common cause of sporadic viral encephalitis in adults)

Common Viral CNS Infection Etiologies

- Enteroviruses:
 - Coxsackievirus A and B
 - Echovirus & Enterovirus
- Arboviruses:
 - Eastern and Western equine encephalitis virus
 - LaCrosse encephalitis virus
 - St. Louis encephalitis virus
- Herpes Simplex Virus (HSV)
- Varicella-zoster virus (VZV)

Less Common Viral CNS Infection Etiologies

- JC Virus: it causes Progressive Multifocal Leukoencephalopathy (PML)
- Rabies virus
- Lymphocytic choriomeningitis virus (LCMV)
- Polio virus: a previous cause of infantile paralysis; preventable with a vaccine
- sIgA is potent aspect of the IR



Acute Viral Meningitis

- Viruses leading cause of aseptic meningitis
- Symptoms:
 - fever,
 - rash,
 - headache,
 - nuchal rigidity,
 - and photophobia, often with
 - an increased number of lymphocytes in CSF (lymphocytic pleocytosis)

Bacterial vs. Viral Meningitis

Acute Meningitis

	Bacterial	Viral
Presentation	Focal or global	Mostly focal
CSF	↑ Opening pressure ↑ PMNs ↑ Protein ↓ Glucose	Normal/↑ Opening pressure ↑ Lymphocytes Normal protein Normal/↑ glucose
Outcomes	High mortality and morbidity	Low mortality and morbidity
Tx	Immediate abx therapy	Supportive therapy

*Note: In infants empirical antimicrobial is initiated even if viral is suspected

Review Question 1

Choose from the options in parentheses to complete the sentences or fill in the blanks:

In a suspected case of meningitis, CSF containing high protein and low glucose may indicate (**bacterial/viral**) acute meningitis which requires _____ treatment.

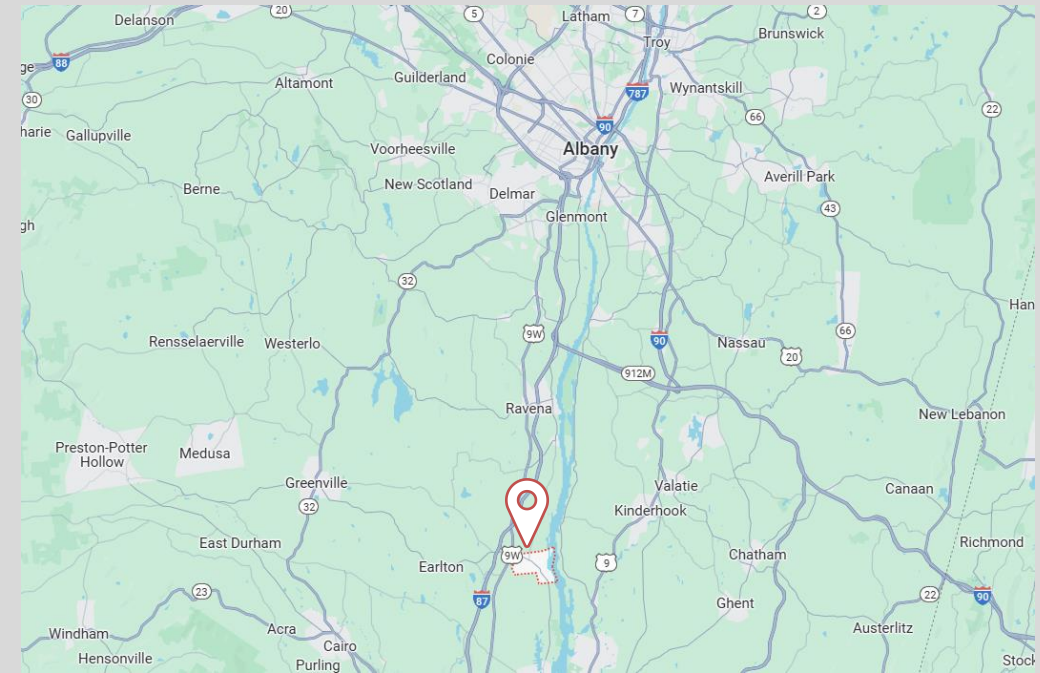
In a suspected case of meningitis, CSF containing high lymphocytes and normal glucose may indicate (**bacterial/viral**) acute meningitis which requires _____ treatment.

Viral Encephalitis

- Encephalitis is usually caused by viruses
- Brain parenchyma
- Encephalitis may have associated meningitis (meningoencephalitis) and, in some cases, involvement of the spinal cord or nerve roots (encephalomyelitis, encephalomyeloradiculitis)
- Sporadic: HSV, VZV and, less commonly, enteroviruses.
- Epidemics of encephalitis are caused by arboviruses, which belong to several different viral taxonomic groups including Alphaviruses (e.g. Eastern equine encephalitis virus, Western equine encephalitis virus), Flaviviruses (e.g., WNV, St. Louis encephalitis virus, and Bunyaviruses (e.g., California encephalitis virus serogroup, LaCrosse virus).

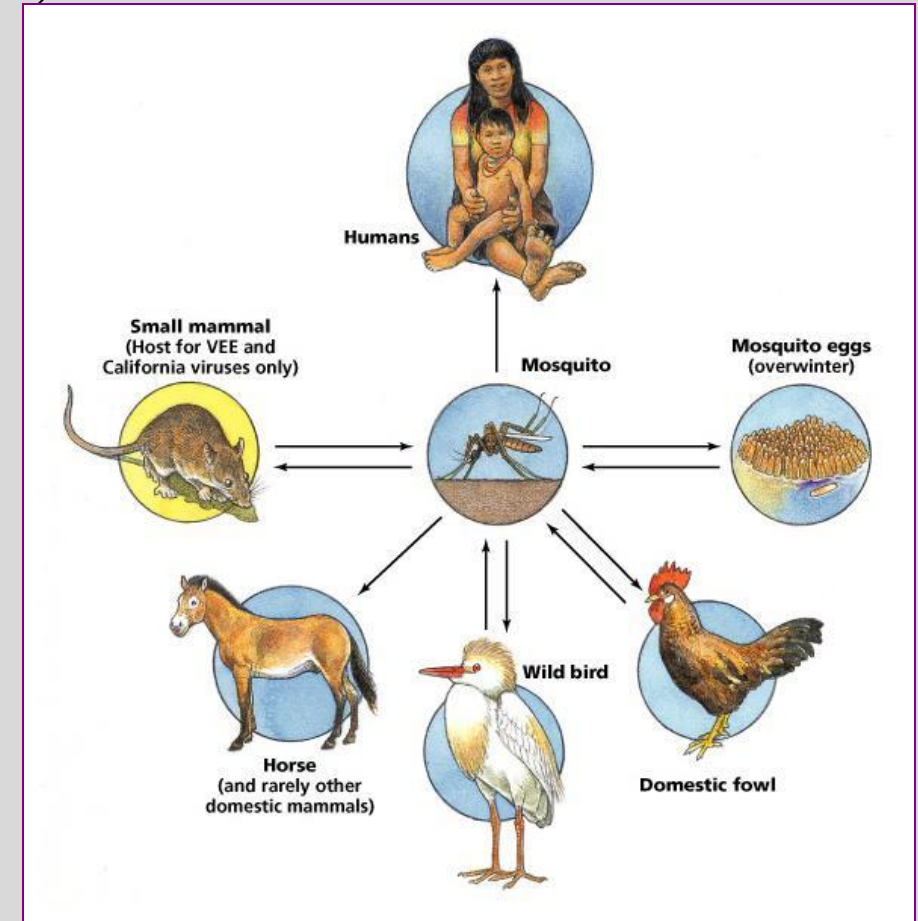
Coxsackieviruses

- Etiologic agent of foot and mouth disease & Herpangina (vesicular diseases)
- Can cause acute viral meningitis
- Diagnosis by NAA
- Disease typically self limiting unless the illness is associated with encephalitis (meningoencephalitis) or occurs in children younger than 1 year



Arboviruses

- Vector-borne illnesses (mostly mosquitoes)
- Examples:
 1. Alphaviruses
 1. Eastern Equine encephalitis virus
 2. Western Equine encephalitis virus
 2. Flaviviruses
 1. St. Louis encephalitis virus
 2. West Nile virus
 3. Japanese encephalitis virus
 3. Bunyaviruses
 1. LaCrosse encephalitis virus
- Most individuals are asymptomatic
- Few develop serious life threatening illness



Arboviruses (EEE)

- Eastern equine encephalitis (EEE) is caused by a virus transmitted to humans and equines (horses) by the bite of an infected mosquito.
- Incubation period: 4-10 days
- Symptoms: sudden onset of fever, myalgia, and a headache of increasing severity
- 1/3 with clinical encephalitis caused by EEE will die from the disease
- Endemic in the New England states

Arboviruses (WEE)

- Western equine encephalitis (WEE) virus is an important cause of encephalitis in horses and humans in North America, mainly in western parts of the USA and Canada.
- The life cycle of WEE involves birds, in which the infection is inapparent, and mosquitoes.
- Symptoms: sudden onset with fever, headache, nausea, vomiting, anorexia and malaise, followed by altered mental status, weakness and signs of meningeal irritation

Arboviruses (SLE)

- In the US, the leading cause of epidemic Flavivirus encephalitis is St. Louis encephalitis (SLE) virus
- SLE is the most common mosquito-transmitted human pathogen in the U.S
- SLE virus is distributed throughout the lower 48 states.
- Since 1964, there have been ~5000 confirmed cases of SLE with an average of 193 cases per year (range 4 - 1,967)
- Illness ranges in severity from a simple febrile headache to meningoencephalitis, with an overall case-fatality ratio of 5-15 %

HSV encephalitis

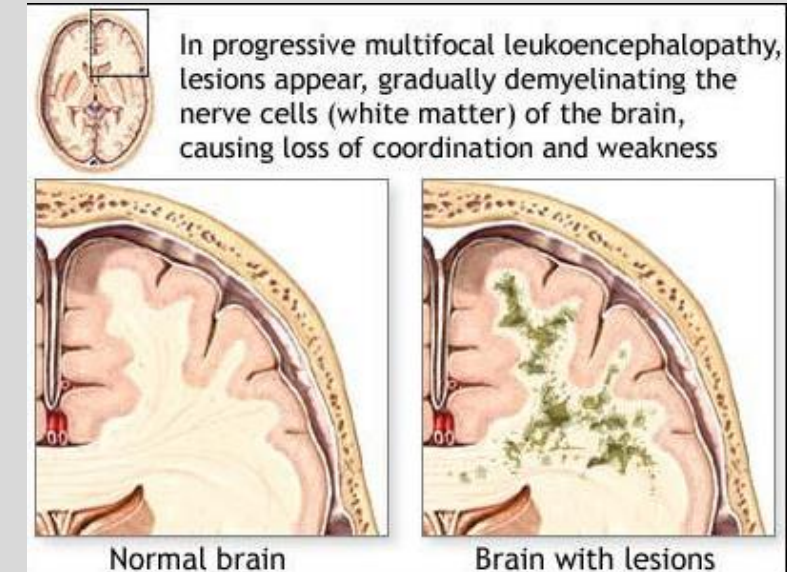
- Adult and pediatric (post-neonatal) HSV encephalitis is usually caused by HSV-1; HSV-1 is the most common viral cause of sporadic viral encephalitis
- Reactivated HSV can spread either to the skin, along the branches of the trigeminal nerve, causing sores on the lips (herpes labialis), or to the brain, infecting the meninges
- Patients often display bizarre behavior, personality changes, insomnia, and gustatory hallucinations
- 70% of neonatal HSV infections are caused by HSV-2 (herpes genitalis), which is acquired during vaginal delivery by contact with secretions in the infected birth canal
- Untreated, neonatal HSV infection can disseminate to the CNS in most cases
- Antiviral therapy: acyclovir and its derivatives, valacyclovir, penciclovir and famciclovir

Varicella-zoster virus (VZV)

- Varicella-zoster virus causes chicken pox & herpes zoster or shingles in reactivation
- Chicken pox is acquired by inhalation but also by contact with skin vesicles
- Virus replicates in the respiratory tract then disseminates via bloodstream to skin producing the characteristic fluid filled vesicles of chickenpox
- The virus then travels from the skin up sensory nerves by retrograde axonal transport and becomes latent in the nerve cell bodies of the dorsal root or cranial nerve ganglia
- Very similar to HSVs (latency, reactivation, neurotropism)
- Zostavax (>60yrs) & Shingrix vaccines (> 50 yrs)

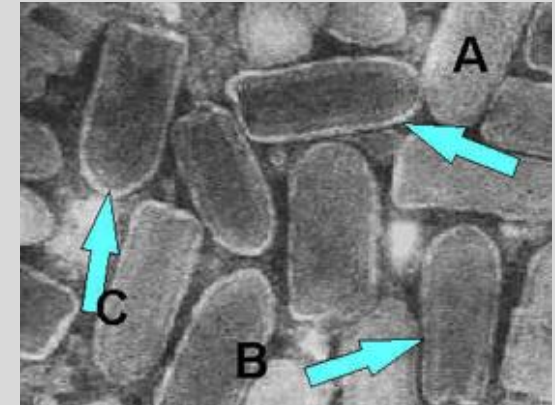
JC virus

- JC virus is a human polyoma virus nonenveloped, ds DNA virus with an icosahedral capsid
- The JC virus enters through the respiratory tract and then spreads to the kidneys
- B cells and monocytes are infected (latency)
- Reactivation occurs in immunocompromised patients where the virus spreads through the blood to CNS (Progressive multifocal leukoencephalopathy; PML)
- PML is a subacute demyelinating & necrotic disease.
- The disease progresses to include paralysis of the arms and legs and death.



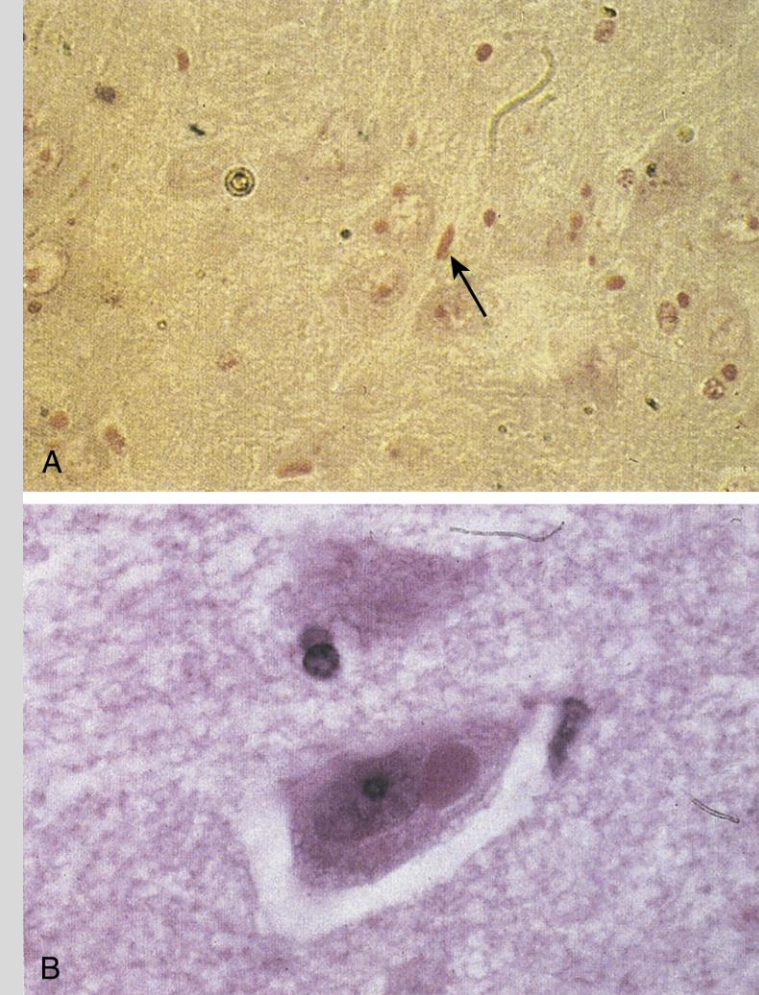
Rabies Virus

- Rhabdoviridae; Bullet-shaped, enveloped, -ve sense ssRNA virus
- Rabies virus causes an acute encephalitis in all warm-blooded hosts, including humans, and the outcome is almost always fatal (Zoonosis)
- Although all species of mammals are susceptible to rabies virus infection, only a few species (esp. racoons & bats) are important as reservoirs for the disease
- Transmission is through a bite containing virus-containing saliva of an infected animal
- After uptake into peripheral nerves, virus transported to CNS via retrograde axoplasmic flow
- During the period of cerebral infection, the classic behavioral changes associated with rabies develop



Rabies Virus

- Rabies infection frequently causes cytoplasmic eosinophilic inclusion bodies (**Negri bodies**) in neuronal cells, especially pyramidal cells of the hippocampus and Purkinje cells of the cerebellum
- Ag detection using direct immunofluorescence
- RT-PCR from multiple samples are needed to confirm the infection (serum, CSF, skin)



Negri bodies caused by rabies. (A) A section of brain from a patient with rabies shows Negri bodies (arrow) . (B) Higher magnification from another biopsy specimen.